# PROJETO 10 - CONTROLE DE SISTEMA TORCIONAL COM DOIS DISCOS

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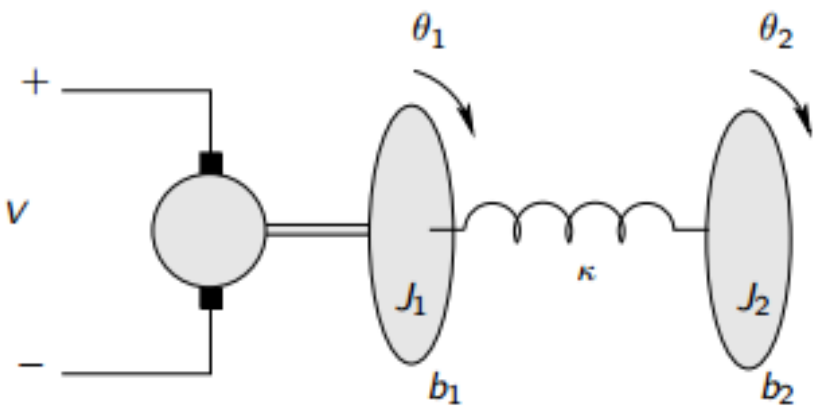
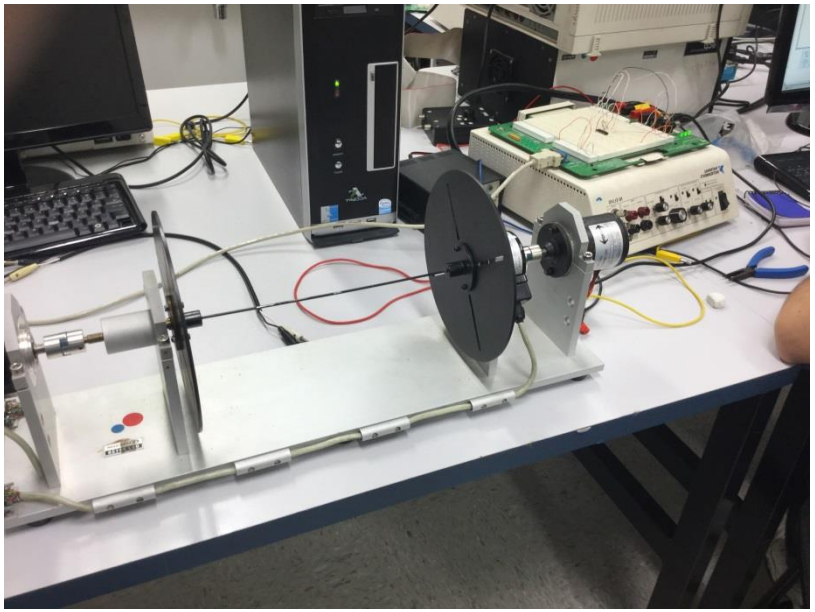
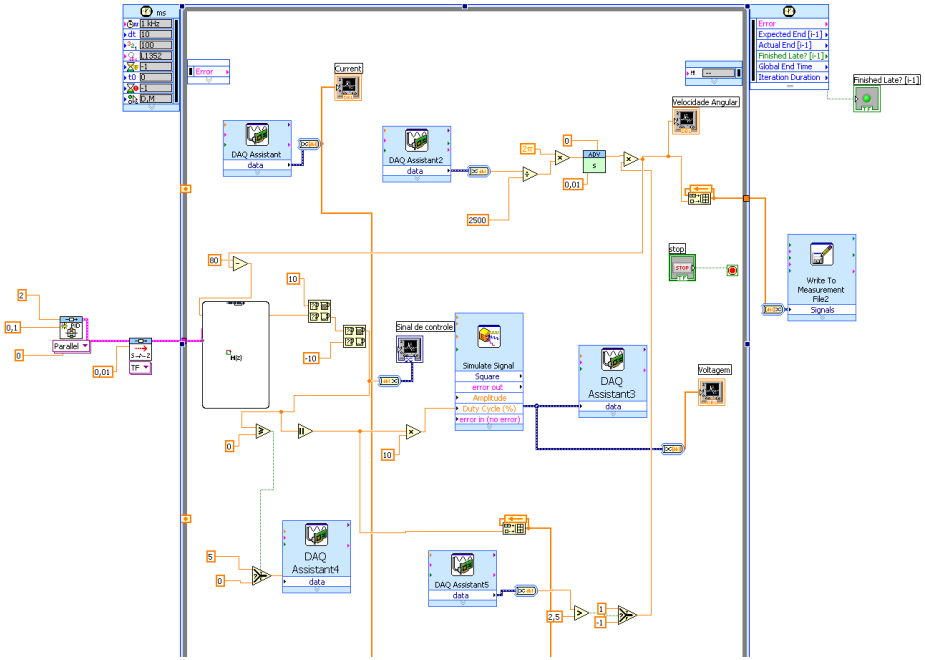
# ESTUDO DA PLANTA

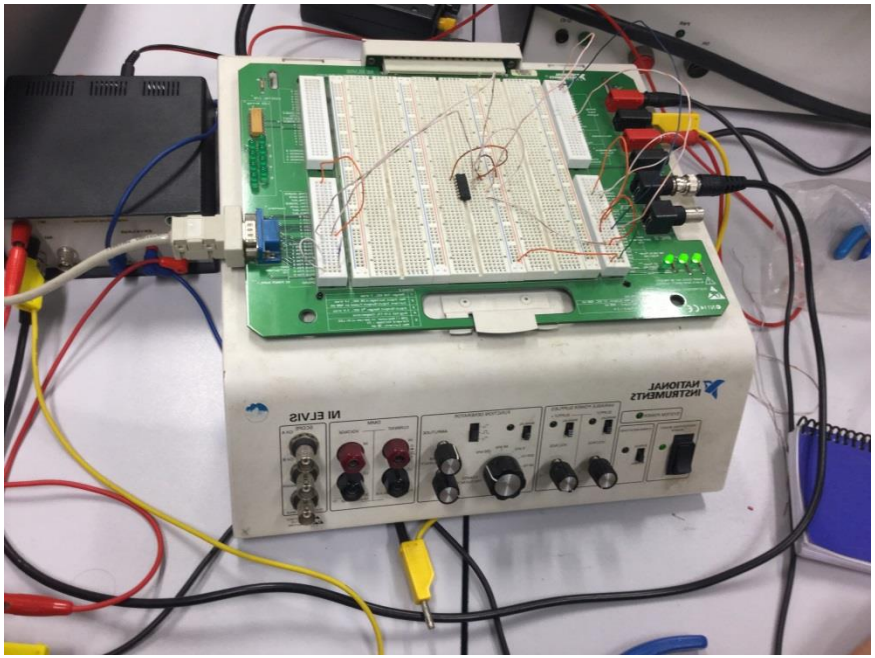
- Parte mecânica
- Parte elétrica



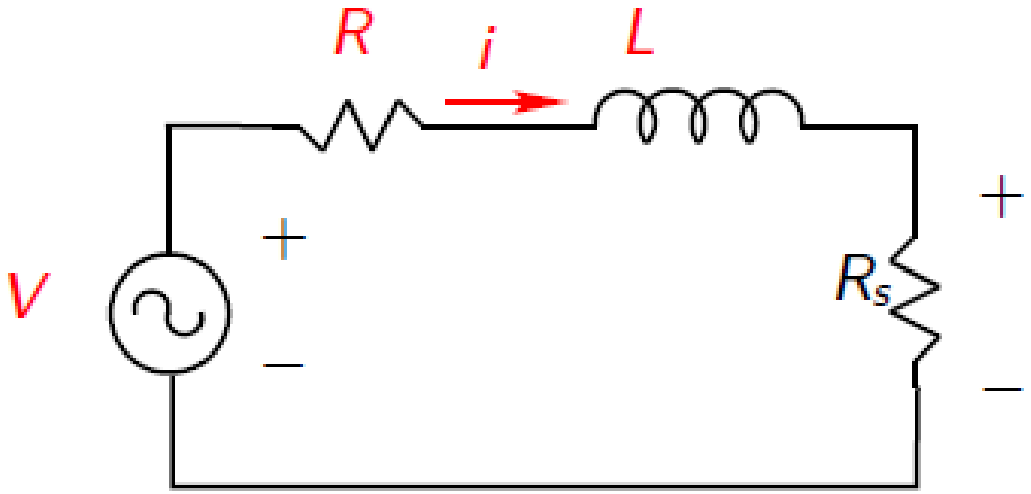
# LABVIEW


Error
Current
DAQ Assistant
data
DAQ Assistant2
data
Velocidade Angular
Finished Late? [i-1]
Stop
Write To Measurement File2
Signals
Sinal de controle
Simulate Signal
Square
error out
Amplitude
Duty Cycle (%)
error in (no error)
DAQ Assistant3
data
Voltagem
DAQ Assistant4
data
DAQ Assistant5
data
Parallel
TF



# PARTE ELÉTRICA

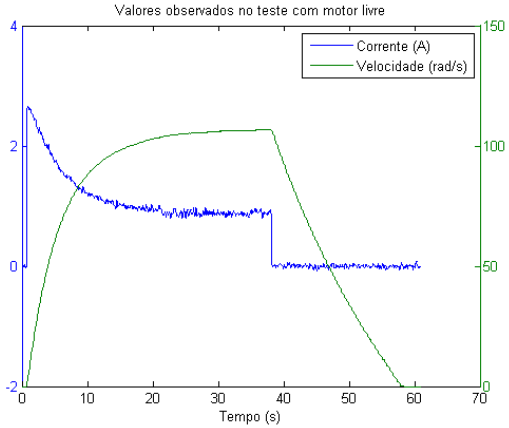
- Estudo com discos travados
- R=1Ω
- L=0H

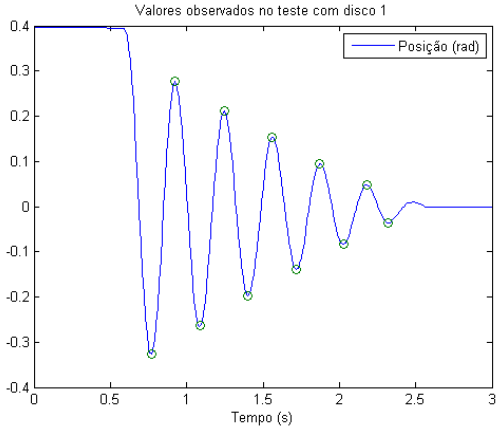


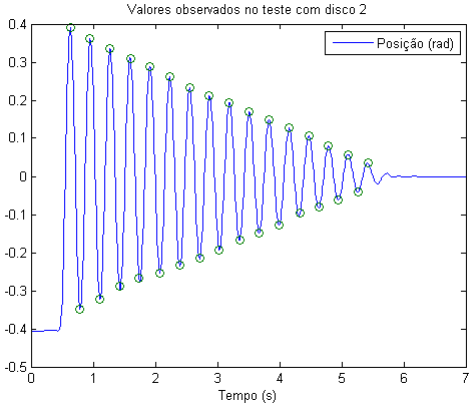

Cálculo das variáveis elétricas

# PARTE MECÂNICA

- 3 estudos
  - 1- 2 discos rodando livremente
  - 2- disco 1 travado
  - 3- disco 2 travado




Calculo das variáveis mecânicas

# PLANTA

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & -1 \\ -\frac{\kappa}{J_1} & -\frac{b_1+K^2/R}{J_1} & 0 \\ \frac{\kappa}{J_2} & 0 & -\frac{b_2}{J_2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{K}{RJ_1} \\ 0 \end{bmatrix} \begin{bmatrix} V \end{bmatrix}$$

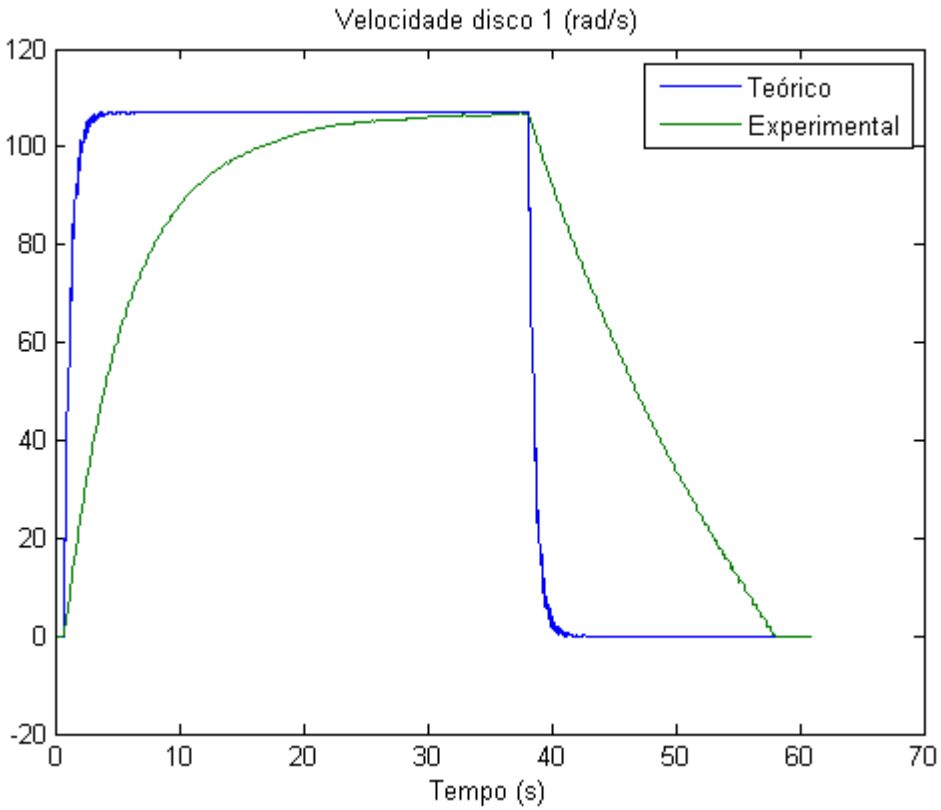
$$\mathrm{b}_1 = 1{,}0651 * 10^{-4}$$
$$\mathrm{b}_2 = 5{,}1545 * 10^{-4}$$
$$k = 0{,}1989$$
$$\mathrm{J}_1 = 4{,}8176 * 10^{-4}$$
$$\mathrm{J}_2 = 5{,}1345 * 10^{-4}$$
$$R = 4{,}4581$$

# PLANTA

- Valores substituídos:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & -1 \\ -412{,}9 & -2{,}875 & 0 \\ 387{,}4 & 0 & -1{,}004 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 35{,}15 \\ 0 \end{bmatrix} * V$$

# COMPARAÇÃO




**Planta teórica muito diferente!**

# ESTRUTURA DO CONTROLADOR

- Controlador PI

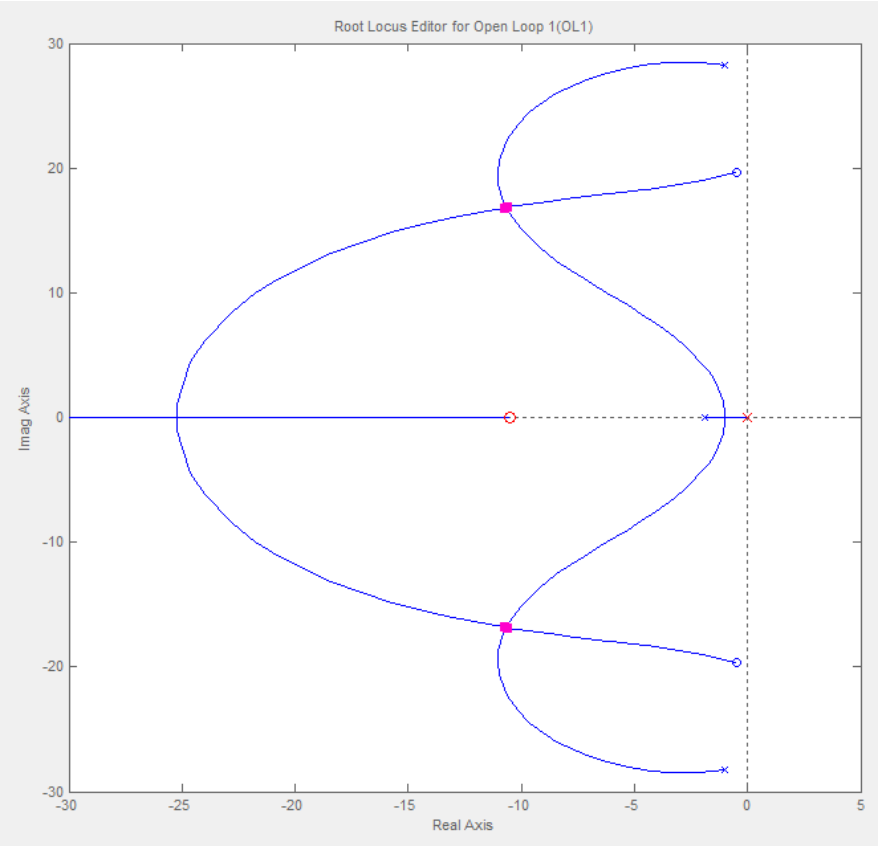
$$C(s) = K_p + \frac{K_i}{s} = \frac{K_p * \left(s + K_i/K_p\right)}{s}$$

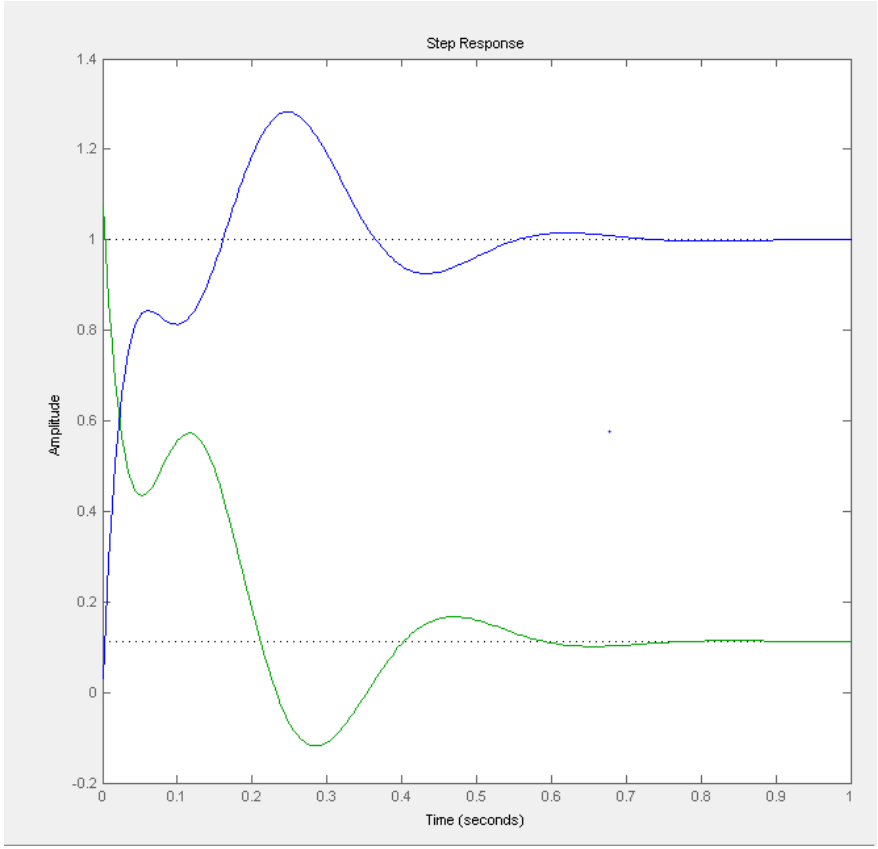
Controlador robusto

# CONTROLE PLANTA TEÓRICA

- Controle PI pelo Sisotool

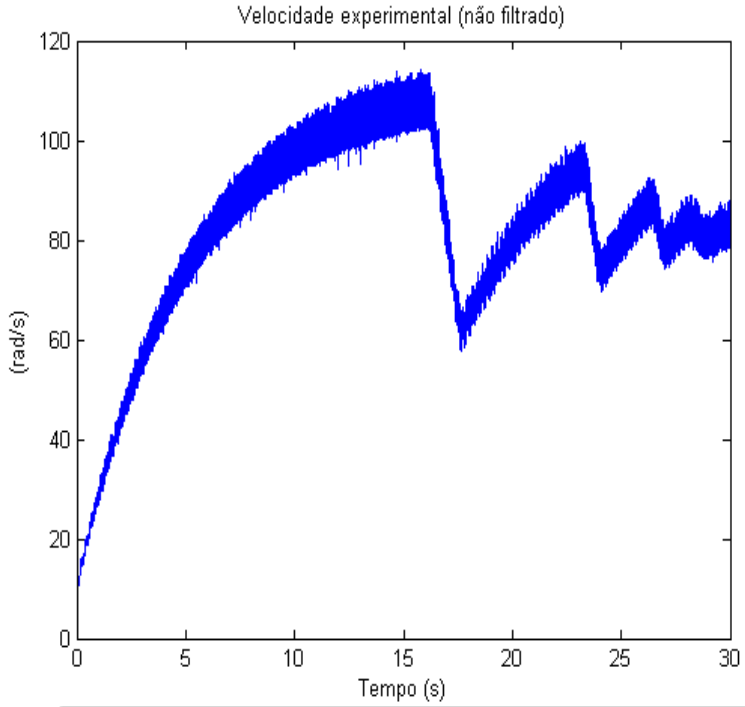
$$C_{siso}(s) = 1.1 + \frac{11.6}{s}$$

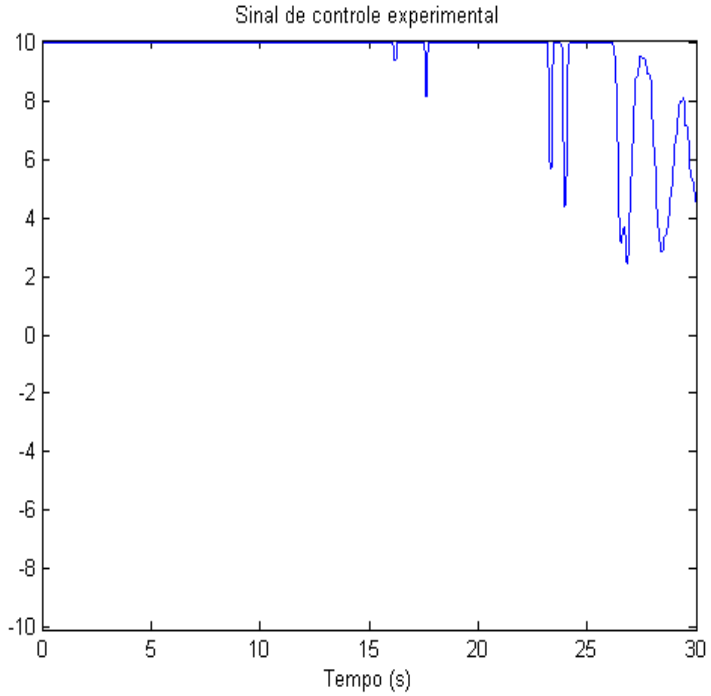


# CONTROLE PLANTA TEÓRICA

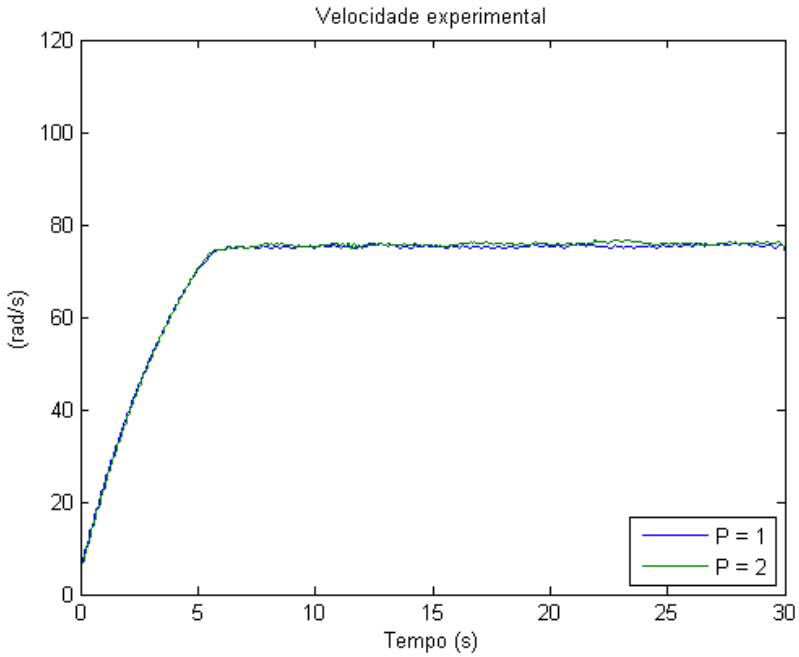
- Resultado prático

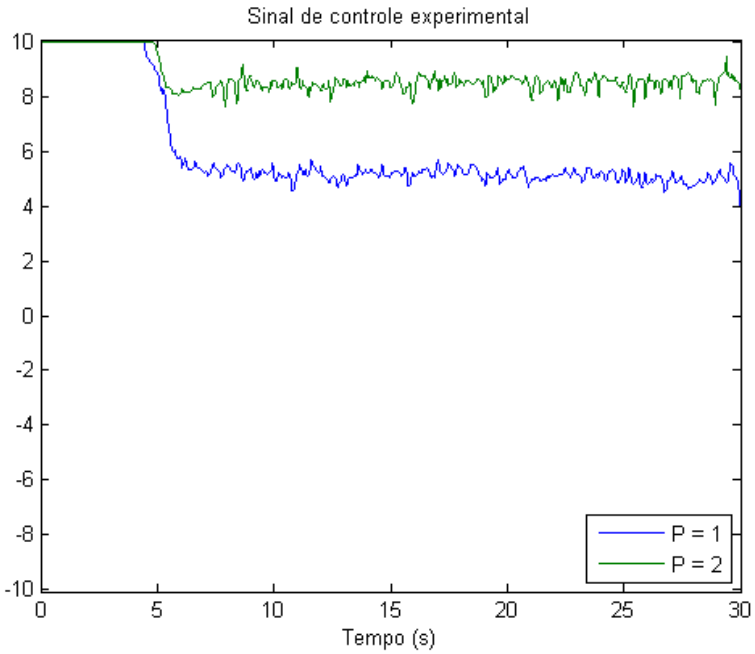



**Controle na planta teórica não condizente**

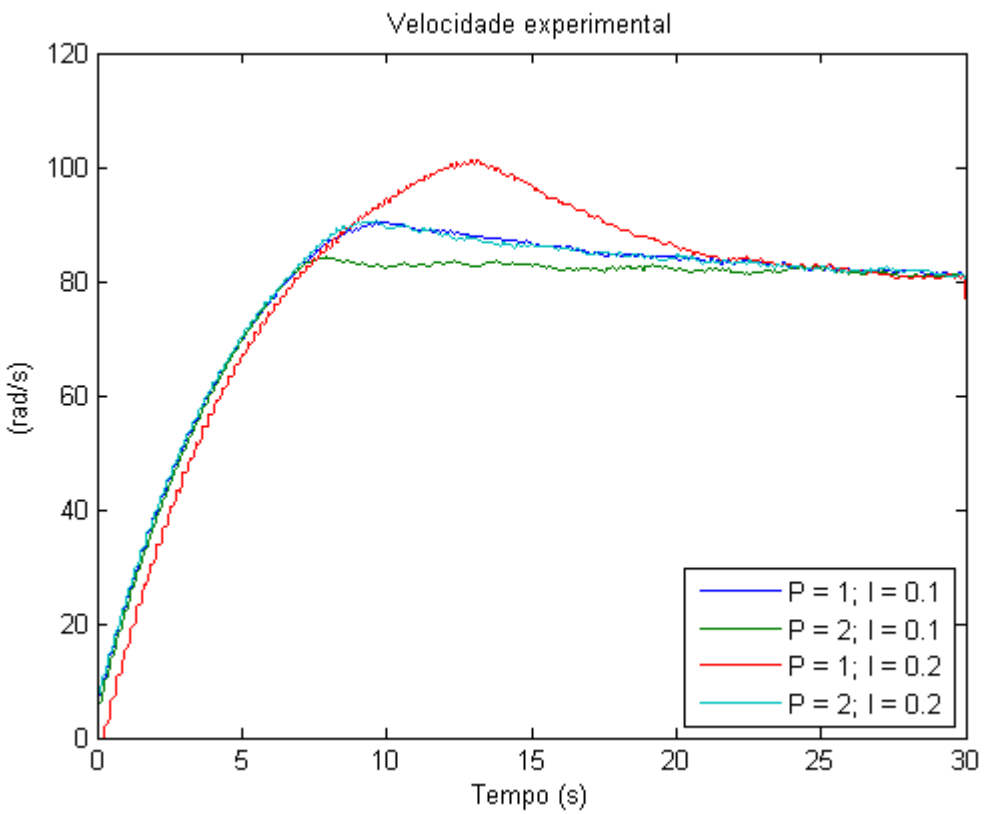
# CONTROLE PROPORCIONAL




Não segue o degrau

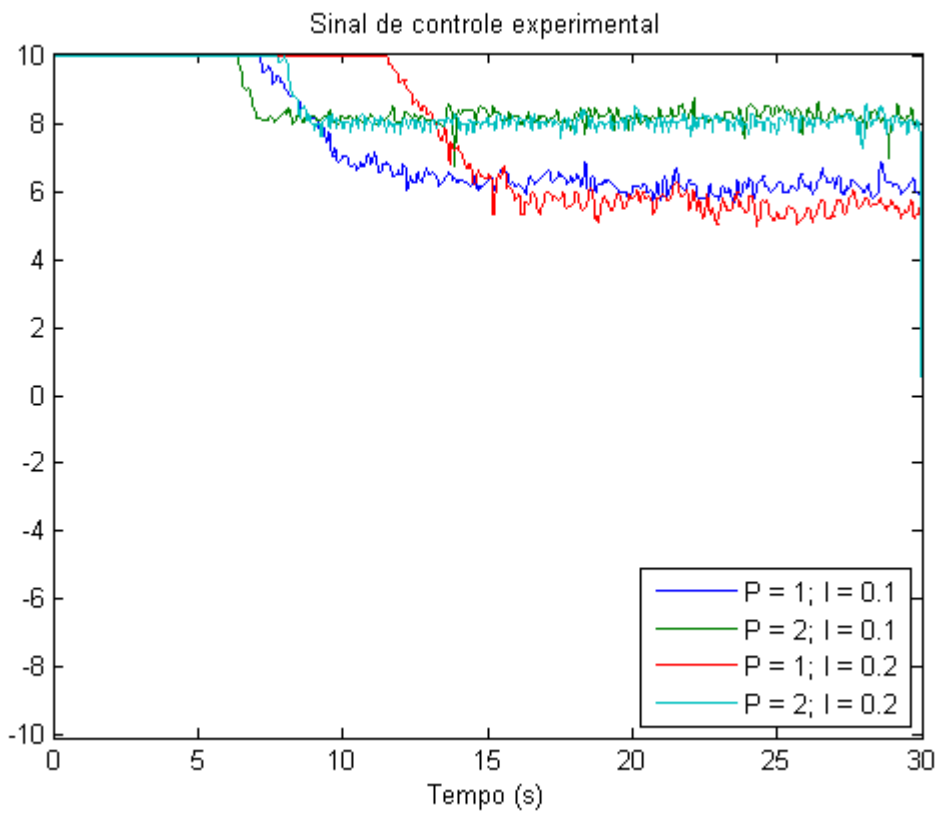
# CONTROLADOR PI




$$C_{verde}(s) = 2 + \frac{0.1}{s}$$

**Parâmetros do verde cumprem os requisitos:**
**Tempo de estabilização – 5,3s**
**Sobressinal – 5,4%**

# CONTROLADOR PI




Sinal de controle satura nos 6s iniciais

Perguntas?